# Free-fall lab

**antocciphysics**



**Name:** _____ **Block:** _____ **Mastery Rating:** _____ (teacher only)

**LEARNING TARGET** Estimate g from measured motion and discuss measurement limits.

**SUCCESS CRITERIA - I can...**

1. Keep supplied examples separate from measurements.
2. Find acceleration from a velocity-time slope.
3. Use repeated trials and explain uncertainty.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “slope” and “trial” to explain your model or result. Sentence starters are optional.

**slope**: vertical change divided by horizontal change on a graph

**trial**: one run or repetition of a test

**acceleration**: rate of change of velocity

## 01 Warm-up: bring back an earlier idea

Two equal 3000 kg masses are 5 m apart.
**Use** $G=6.67\times10^{-11}$ N·m²/kg².
**Find** force. By what factor is an answer wrong if r rather than r² is used?

**Sentence starter:** The step I would change is … . The evidence is … .

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $a = \Delta v/\Delta t$; $v_f = v_i + a\,\Delta t$.
Velocity-time slope estimates acceleration. Downward is positive. Use the approved apparatus; distinguish supplied from measured data.

**Worked reasoning:** Supplied trial values 8, 10, 12 m/s² have mean 10 m/s² and range 8-12. The spread describes variation; averaging does not remove every systematic error.

**Your turn:** Plot the three values on a number line and mark the mean.
**Name** an error that could shift all trials together.

**Sentence starter:** The step I would change is … . The evidence is … .

| Trial | Estimated g (m/s²) |
|---|---|
| 1 | 8 |
| 2 | 10 |
| 3 | 12 |

**Marzano check - circle:** ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

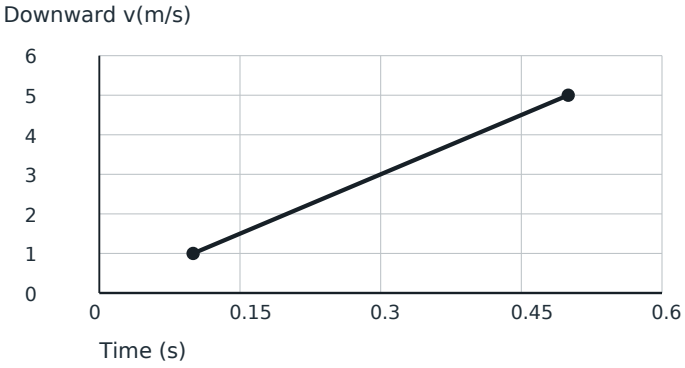
Sentence starters are optional. You may plan in any language.

## 03 Read and interpret the model

**Use** the supplied velocity-time graph to estimate acceleration.
**Show** the change in velocity and elapsed time.

**Word help:** estimate = find an approximate value

**Sentence starter:** I found … [unit]. This value means … in this situation.




**Marzano check - circle:** ① ② ③

## 04 Apply the physics

Three video trials estimate g as 9.6, 9.9 and 10.2 m/s².
**Find** the mean and half-range.
**Compare** the mean with 9.8 m/s² using percent difference from that reference.
**Label** a sketch of the measurement interval.

**Word help:** estimate = find an approximate value

**Sentence starter:** Both … . They differ in … because … .

| Trial | Estimated g (m/s²) |
|---|---|
| 1 | 9.6 |
| 2 | 9.9 |
| 3 | 10.2 |

**Marzano check - circle:** ① ② ③

## 05 Explain your reasoning

Plan three repeated trials.
**Name** a controlled condition, a measurement you will record and a reason not to include a badly pushed release without comment.

**Sentence starter:** My design shows … . It meets the requirement by … .

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 06 Review: connect an earlier idea

A 4 kg lamp hangs at rest from a vertical rope.
**Use** g=9.8.
**Find** tension. Then find tension if it accelerates upward at 3 m/s².
**Draw** the forces on the chosen object and label their sources.

**Word help:** tension = the pulling force through a stretched rope or cable

**Sentence starter:** I found … [unit]. This value means … in this situation.

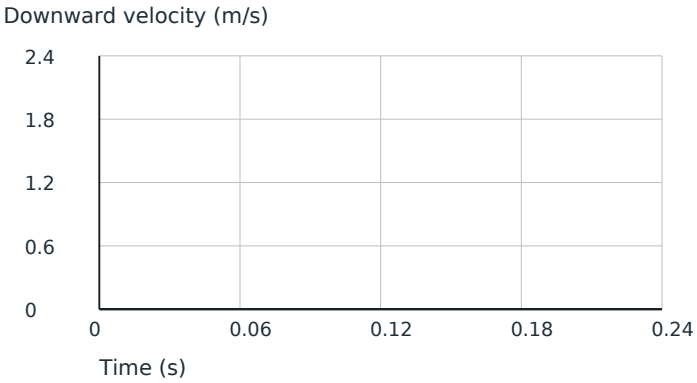
Add and label force arrows.

**Marzano check - circle:** ① ② ③

## 07 Exit check

A video gives downward velocities of 0.15 m/s at 0.02 s and 2.11 m/s at 0.22 s. Plot the points and find the slope.
**Explain** why fitting several reliable measurements can be better than using one very short interval.

**Sentence starter:** Under the assumption …, the evidence … supports … because … .




**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the model

Supplied estimates are 9.4, 9.8, 10.2 m/s².
**Find** mean, half-range and percent deviation of mean from 9.81. Could zero spread guarantee accuracy?

**Sentence starter:** I found … [unit]. This value means … in this situation.

| Group / trial | Values (m/s²) | Mean | Spread |
|---|---|---|---|
| 1 | ____ | ____ | ____ |
| 2 | ____ | ____ | ____ |

**Marzano check - circle:** ① ② ③